

Jamhuriya RMS

System Documentation

Research Management System — Full Guide

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Stack: MERN (MongoDB, Express, React, Node.js)
See also: docs/DATABASE_STRUCTURE.pdf

1. System Overview

Jamhuriya RMS is a web-based research management platform covering the full research lifecycle: proposal submission, ethics review (REC), project tracking, grants and budgets, publications, institutional repository, collaboration, and analytics.

Architecture: React SPA (Vite) !' Express REST API !' MongoDB. JWT authentication with HTTP-only refresh cookies. Role-based access control (RBAC) on both frontend routes and backend endpoints.

2. Technology Stack

Layer	Technology	Notes
Frontend	React 19 + Vite	Port 5173, proxies /api to backend
Backend	Express 5 + Node.js	Port 5000
Database	MongoDB + Mongoose	Database: rms
Auth	JWT + bcrypt	Access token + refresh cookie
Charts	Recharts	Dashboard analytics
PDF	PDFKit	Reports, ethics certificates, exports
Files	Multer	Proposal docs, repository uploads

3. User Roles

Role	Code	Responsibilities
Research Director	research_director	Institutional oversight, ethics approval, grants, user management, departments, analytics Faculty proposal pre-review, publication validation, faculty reports
Faculty Coordinator	faculty_coordinator	
Finance Officer	finance_officer	Budgets, payment disbursement, procurement, finance reports
Researcher	researcher	Proposals, ethics forms, projects, publications, groups, repository uploads

4. Frontend Pages

Route	Page	Access
/dashboard	Dashboard & Analytics	All roles — role-specific widgets
/ethics	Ethics (REC)	Director, Coordinator, Researcher
/proposals	Proposals	Director, Coordinator, Researcher
/proposals/new	New Proposal + Ethics	Researcher only
/proposals/:id/review	Director/Coordinator Review	Director, Coordinator
/projects	Projects	Director, Coordinator, Researcher
/publications	Publications & Outputs	Director, Coordinator, Researcher
/research-workflow	Research Workflow	Director, Coordinator, Researcher
/thesis	Thesis Groups	Director, Coordinator, Researcher
/grants	Grants	All except standalone coordinator-only
/budgets	Finance & Budgets	Director, Finance, Researcher
/repository	Repository	Director, Coordinator, Researcher
/collaboration	Collaboration Hub	All authenticated roles
/groups	Research Groups	Director, Coordinator, Researcher
/messages	Messages	All authenticated roles
/notifications	Notifications	All authenticated roles
/pending-users	User Approval	Director only
/departments	Departments CRUD	Director only
/profile	Profile	All roles

5. REST API Endpoints

Base URL: <http://localhost:5000>. All routes except `/api/auth/register`, `/api/auth/login`, `/api/health`, and `/api/repository/oai` require authentication.

`/api/auth`

- POST `/register` — self-registration (pending approval)
- POST `/login` — JWT access + refresh cookie
- POST `/logout` — clear session
- POST `/refresh` — renew access token
- GET `/me` — current user profile

`/api/users`

- GET `/` — list users (director)
- GET `/pending` — pending registrations (director)
- POST `/:id/approve` | `/:id/reject` — user approval
- PUT `/me` — update own profile

`/api/proposals`

- GET `/` — list proposals (role-filtered)
- POST `/` — create proposal (researcher, multipart document)
- PUT `/:id` — update draft/revision (researcher)
- POST `/:id/submit` — submit proposal + linked ethics to director
- POST `/:id/review` — coordinator recommendation
- POST `/:id/director-decision` — approve/reject (creates project on approve)
- GET `/:id/ethics-application` — linked ethics form

`/api/ethics`

- GET `/` — list ethics applications
- POST `/` — create standalone ethics application
- PATCH `/:id` — save draft
- POST `/:id/submit` — submit to REC
- POST `/:id/director-decision` — approve/reject + issue certificate
- GET `/:id/certificate.pdf` — download ethics certificate

`/api/projects`

- GET `/` — list projects
- GET `/:id` — project details with PI
- PUT `/:id` — update project (director/researcher)
- POST `/:id/progress` — add progress report (researcher)

`/api/publications`

- GET `/` — list publications
- POST `/` — create publication (researcher)
- PUT `/:id` — update draft
- POST `/:id/submit` — submit for validation
- POST `/:id/validate` | `/:id/reject` — coordinator/director validation
- GET `/citation?doi=` — CrossRef DOI lookup

`/api/grants`

- GET `/` — list grants
- POST `/` — create grant request (researcher)
- PUT `/:id` — update grant
- POST `/:id/submit` — submit for review
- POST `/:id/director-decision` — approve with awarded amount

/api/budgets

- GET / — list budgets
- POST / — create budget (director/finance)
- POST /:id/items — add budget line item
- PATCH /:id/items/:itemId — approve/reject/pay item

/api/payments

- GET / — list payment requests
- POST / — create payment request
- POST /:id/director-decision — director approve/reject
- POST /:id/finance-pay — finance marks as paid

/api/procurement

- GET / — list purchase orders
- POST / — create PO
- POST /:id/director-decision — director approve/reject
- POST /:id/finance-pay — finance marks as paid

/api/repository

- GET / — list repository items
- POST / — upload file (PDF/CSV/Excel only)
- GET /export/csv | /export/pdf | /export/excel — catalog export
- GET /oai — OAI-PMH 2.0 (public, app-level route)
- GET /oai/export — OAI bulk export

/api/groups

- GET / — list research groups
- GET /stats — collaboration vs thesis counts
- POST / — create group
- POST /:id/join | /:id/leave — membership
- DELETE /:id — delete group (director)

/api/thesis-groups

- GET / — list thesis groups
- POST / — create thesis group
- PATCH /:id — update group
- POST /:id/meetings — log supervision meeting
- DELETE /:id — delete (director)

/api/conversations

- GET / — list my conversations
- GET /users — messageable users list
- POST / — start 1:1 conversation
- GET /group/:groupId — open group chat
- POST /:id/messages — send message (+ notification)

/api/notifications

- GET /me — my notifications
- GET /me/unread-count — badge count
- POST /:id/read — mark as read

/api/analytics

- GET /dashboard — role-based dashboard metrics
- GET /institutional — director institutional analytics
- GET /faculty-report — coordinator faculty report

- GET /faculty-report.pdf — faculty report PDF export
- GET /finance-report — finance summary
- GET /annual-report.pdf — director annual report PDF

/api/departments

- GET / — list departments
- POST / — create (director)
- PUT /:id — update (director)
- DELETE /:id — delete (director)

6. Key Workflows

A. Proposal + Ethics (Researcher != Director)

1. Researcher creates Proposal + Ethics form on one page (/proposals/new).
2. System auto-fills ethics from profile and proposal fields.
3. Researcher saves draft or submits both to Director in one action.
4. Director reviews ethics certificate + proposal on /proposals/:id/review.
5. On approval: Project is created automatically; notifications sent.

B. Publications (Researcher != Coordinator/Director)

1. Researcher creates publication record (draft).
2. Submits for validation; appears in faculty workflow pipeline.
3. Coordinator or Director validates or rejects.
4. Validated publications count toward faculty analytics.

C. Finance (Researcher != Director != Finance)

1. Grant approved with awarded amount != Budget created.
2. Researcher requests Payment or Purchase Order against budget.
3. Director approves request.
4. Finance Officer marks payment as paid / disbursed.

D. Collaboration & Messaging

1. Users create or join Research Groups (/groups).
2. Group chat opens via Conversation linked to groupId.
3. Direct 1:1 messaging via /messages with user picker.
4. Each message triggers a Notification for recipients.

E. Thesis Supervision

1. Coordinator/Director creates Thesis Group with students + supervisor.
2. Optional link to ResearchGroup for chat.
3. Supervisor logs meetings (date, location, agenda, notes).
4. Status tracked: proposed != in_progress != submitted != defended != completed.

7. Database (MongoDB)

Database name: rms | URI: mongodb://localhost:27017/rms | Collections (15):

users, departments, proposals, ethicsapplications, projects, publications, grants, budgets, payments, purchaseorders, researchgroups, thesisgroups, conversations, notifications, repositoryitems

Full field-level documentation: docs/DATABASE_STRUCTURE.pdf (generate with npm run docs:database-pdf).

8. Demo Accounts (after npm run seed)

Role	Email	Password
Research Director	director@rms.edu	Director2024!

Faculty Coordinator	coordinator@rms.edu	Coordinator2024!
Finance Officer	finance@rms.edu	Finance2024!
Researcher	asha@rms.edu	Researcher2024!
Researcher	mahad@rms.edu	Researcher2024!

9. Setup & Run

Prerequisites: Node.js 18+, MongoDB running locally.

Backend: `cd backend !' cp .env.example .env !'`
`npm install !' npm run seed !' npm run dev`

Frontend: `cd frontend !' npm install !' npm run dev`

App URL: `http://localhost:5173` | API: `http://localhost:5000/api/health`

10. Export Features

- Repository catalog: PDF, Excel, CSV (/repository page buttons)
- Ethics certificate: PDF per approved application
- Faculty research report: PDF for coordinators
- Annual institutional report: PDF for director
- OAI-PMH 2.0: `/api/repository/oai` (institutional repository integration)

